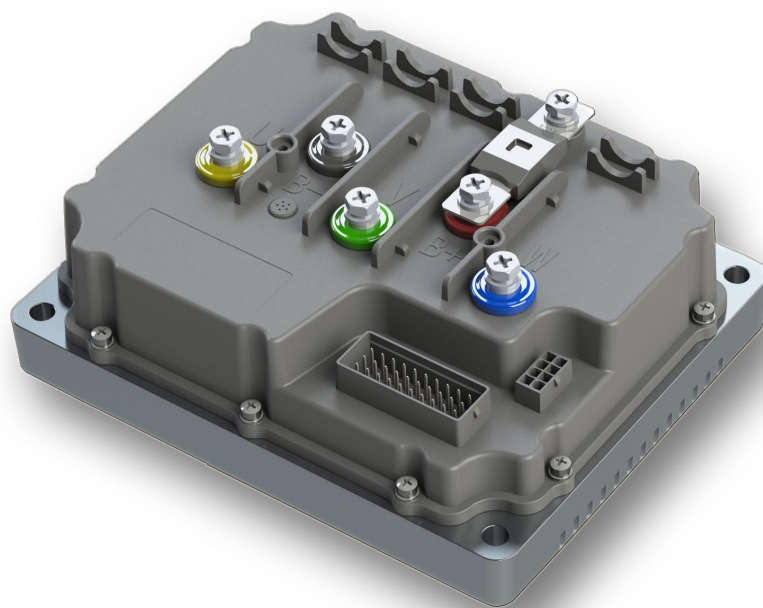




MINDOCOM

永磁同步电机控制器说明书

型号： FDK3533C-XC Three-wheeled version



— Leader of Motor Control

Foreword

Thank you for your support of Wind Control.FDKWith the support and trust of our series of servo drives, you will enjoy our comprehensive and sincere service.

This instruction manual provides you withFDK3533C-XCThis manual contains safety information, mechanical and electrical installation instructions, controller port descriptions, and commissioning and maintenance guidelines. Please read this manual carefully before use and keep it safe. If you encounter any problems during use, please consult our technical support personnel for assistance.

As we are committed to continuous improvement of the controller, the instruction manuals provided will also be updated in a timely manner.

Version Change Log

date	Revised version	Changes
2023Year10moon	V1.0	First edition released

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1 Security Information

1.1 Security Statement

- To ensure personal and equipment safety, please read and comply with this safety information before installing, operating, and maintaining the product. The safety warnings in this manual do not represent all safety precautions and are only supplementary to these safety guidelines. This product should be used in an environment that meets the design specifications; otherwise, malfunctions may occur. Functional abnormalities or component damage caused by failure to comply with relevant regulations are not covered by the product warranty.
- Fengdekong shall not be liable for any damage caused by the following circumstances:
 - 1 Failure to follow operating instructions;
 - 2 Modifying the controller without permission or repurposing the controller for special applications, such as lifting equipment;
 - 3 Improper operation of the controller;
 - 4 Natural damage such as floods and lightning strikes;
 - 5 Damage caused during transportation.

1.2 Electrical safety

The voltage applied to the controller can cause severe electric shock or burns, endangering personal safety. Extreme caution should be exercised when operating or approaching the controller. See the relevant section of the user manual for specific warnings.

1.3 Safety Guidelines

System design, installation, commissioning, and maintenance must be performed by trained and experienced engineers. This safety information and user manual should be read carefully before operation.

The controller's stop and safety functions do not guarantee the isolation of input and output voltages, which could cause personal injury. Therefore, the power supply must be disconnected with a qualified electrical insulation device before making any circuit connections.

In any case, if the controller fails, a hazard analysis must be performed, and if necessary, further measures must be taken to reduce the hazard. For example, if the speed control fails, an overspeed protection device should be used, or if the motor brake fails, a fault-protected mechanical brake device should be used.

1.4 Environmental requirements

The transportation, storage, installation, and use of the controller must comply with the relevant instructions in this user manual and specific environmental requirements.

1.5 Operating Instructions

The design, installation, and commissioning personnel shall comply with all relevant regulations, such as national wiring regulations, accident prevention regulations, and electromagnetic compatibility (EMC) regulations. The selection of cables, fuses, and other protective devices shall conform to relevant design selection specifications, and the equipment shall be reliably grounded.

1.6 Motor safety

- Ensure that the motor is installed and maintained in accordance with the motor manufacturer's instructions.
- For the first commissioning, the motor must be placed under no-load; otherwise, it may fail to reach zero, causing the motor to stall or run away and damage the equipment.
- The rated current of the driven motor should be below the rated current of the controller. If it exceeds or is too low, please consult our technical support personnel. Before operation, be sure to confirm the maximum current value of the controller, otherwise it may damage the motor.
- The motor may overheat when running at low speed. The motor should be equipped with a thermistor and connected to the controller interface for protection. If necessary, a forced cooling fan should also be installed.
- The motor parameters within the controller can affect the normal operation of the motor, and the default values are not applicable to all motors.
- Do not touch a motor that has been running for an extended period of time, as this may result in burns from high temperatures.

1.7 Parameter adjustment

Certain parameters have a significant impact on controller operation. Before changing such parameters, the impact on the control system must be carefully considered; parameters should not be modified blindly.

1.8 Security level



"Danger" means that failure to follow the procedures could result in death or serious bodily injury.

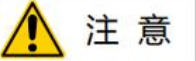


A "warning" indicates that failure to follow instructions may result in death or serious bodily injury.



"Caution" indicates that failure to operate according to regulations may result in minor bodily injury or equipment damage.

1.9 Safety Precautions

Unpacking and inspection	
	- Before opening the box, please check the outer packaging for any damage, moisture, or deformation. Open the box layer by layer, and do not subject the product to any forceful impact! - Upon unpacking, please inspect the product and accessories for any damage, rust, or dents. After unpacking, carefully check the packing list to ensure the quantity and documentation of the product and accessories are complete.
	- If you find any damage, rust, or signs of use on the product or its accessories upon unpacking, please do not install it! Please contact us immediately.

Sales processing.

- If water is found inside the product, or if any parts are missing or damaged upon unpacking, do not install it! Please contact sales immediately for assistance. Please
- carefully check the packing list. If the delivered product does not match the required model, do not install it! Please contact sales immediately for assistance.

Storage and transportation



- Please store and transport the product according to the specified storage and transportation conditions, ensuring the required temperature and humidity. Avoid
- storing and transporting in harsh environments such as strong magnetic fields, strong electric fields, strong vibrations, direct sunlight, water splashes, or rain. Do not
- store the product for more than 3 months. If storage time exceeds this limit, please implement strict protective measures and necessary inspections. Ensure the
- product is properly packaged before vehicle transport; for long-distance transport, sealed containers must be used. Do not transport this product together with
- equipment or items that may affect or damage it.



- Please use professional loading and unloading equipment to move large or heavy equipment and products! When moving products by hand, please hold
- them firmly to prevent them from falling, which could cause injury! When moving products, please lift and place them gently, and always be aware of
- objects under your feet to prevent tripping or falling, which could cause injury or product damage!
- When equipment is lifted by lifting equipment, personnel are prohibited from standing or staying under the equipment.

During installation



- Please read the product instruction manual, safety information, and related precautions carefully before installation! Modification of
- this product is strictly prohibited!
- Do not loosen the fixing screws on the product. If any are found to be loose, tighten them and conduct regular inspections afterwards! Do not
- install this product in places with strong electric fields or strong electromagnetic interference!
- When this product is installed in a cabinet or terminal device, the cabinet or terminal device must be provided with appropriate fireproof enclosures, electrical protective enclosures, and mechanical protective enclosures, and the protection level should comply with relevant IEC standards and local laws and regulations.



- Non-professionals are strictly prohibited from installing, wiring, maintaining, repairing, or replacing parts of this product! Only qualified
- professionals with electrical training and sufficient electrical knowledge may perform the installation, wiring, maintenance, repair, or
- replacement of parts of this product.
- Installers must be familiar with product installation requirements and related technical documentation.
- When installing equipment that is subject to strong electromagnetic interference, such as transformers, please install shielding protection devices to prevent product malfunctions!

When wiring



- Non-professionals are strictly prohibited from installing, wiring, maintaining, repairing, or replacing components on this equipment! Do not perform wiring work while the power is on, as this may result in electric shock. Before wiring, disconnect all equipment power. After disconnecting the power, residual voltage may remain in the internal capacitors; please wait at least 15 minutes before performing any wiring operations. Always disconnect the power first, then test for voltage to ensure there is no current before proceeding. Ensure that all equipment and products are properly grounded; otherwise, there may be a risk of electric shock.
-
- Please follow the procedures specified in the electrostatic discharge (ESD) prevention measures and wear an anti-static wrist strap when performing wiring operations to avoid damaging the internal circuitry of the equipment or product.



- It is strictly forbidden to connect the power input to the output terminal of the equipment or product, otherwise it will cause damage to the equipment or even cause a fire! When connecting the product to the motor, please make sure that the phase sequence of the controller and the motor terminals is accurate and consistent to avoid motor stalling or runaway.
- The cables used in wiring must meet the corresponding requirements for wire diameter and shielding. The shielding layer of shielded cables must be reliably grounded at one end!
- After wiring is completed, ensure that there are no loose screws or exposed cables inside the equipment and product, and carefully check the wiring according to the product interface to avoid reversed polarity, otherwise it may damage the product or motor encoder.

When powered on



- Before powering on, please ensure that the equipment and products are properly installed and the wiring is secure. If it is the first time powering on for matching and debugging, the motor should be placed under no-load and within the safe rotation range.
- Before powering on, please ensure that the power supply meets the equipment requirements and the voltage level is correct to avoid equipment damage or fire! When powering on, the mechanical devices of the equipment or product may suddenly operate. Please be careful to prevent mechanical injury and ensure a sufficient safe distance from the mechanical devices.
- After powering on, do not open the equipment cabinet door or product protective cover, otherwise there is a risk of electric shock! It is strictly forbidden to touch any wiring terminals of the equipment while it is powered on, otherwise there is a risk of electric shock! It is strictly forbidden to disassemble any device or component of the equipment or product while it is powered on, otherwise there is a risk of electric shock!

runtime



- Do not touch any terminals of the equipment while it is running, as this may result in electric shock! Do not disassemble any device or component of the equipment or product while it is running, as this may result in electric shock! Do not touch the equipment casing, fan, motor, or resistors to test the temperature, as this may cause burns!

-Non-professional technicians are strictly prohibited from testing signals during operation, as this may result in personal injury or equipment failure!



警告

- During operation, prevent other objects or metal objects from falling into the equipment, as this may cause
- damage! Do not use contactors to control the start and stop of the equipment, as this may also cause damage!

During maintenance



危险

- It is strictly forbidden for non-professionals to install, wire, maintain, repair, or replace components on this equipment! It is strictly forbidden to
- perform equipment maintenance while the power is on, as this poses a risk of electric shock!
- After disconnecting the power to all equipment, please wait at least 15 minutes before performing any maintenance operations. Always disconnect the power first, then test for any remaining power before proceeding.



警告

-Please perform daily and periodic inspections according to the equipment maintenance and upkeep requirements, and keep maintenance records.

During maintenance



危险

- It is strictly forbidden for non-professionals to install, wire, maintain, repair, or replace components on this equipment! It is strictly forbidden to
- perform any repairs while the equipment is powered on, as this poses a risk of electric shock!
- After disconnecting the power to all equipment, please wait at least 15 minutes before performing any maintenance operations. Always disconnect the power first, then test for any remaining power before proceeding.



警告

- Please report the repair according to the product's warranty agreement process.
- When a product malfunctions or is damaged, you can refer to the relevant troubleshooting manual or consult our technical support personnel for professional troubleshooting and repair, and keep relevant repair records.
- Please replace the parts according to the product's easy-to-wear parts replacement instructions.
- Do not continue to use the damaged product, as this may lead to further damage or even an accident. After replacing the product, please be sure to re-
- confirm and verify the relevant wiring, and calibrate the parameters. The motor should be tested under no-load until it is confirmed to be working properly before using it!

When scrapped



警告

- Please scrap equipment and products in accordance with relevant national regulations and standards to avoid property damage or personal injury! Scrapped equipment
- and products should be treated and recycled in accordance with industrial waste gas treatment standards to avoid environmental pollution.

2 Product Overview

2.1 Product Introduction

FDK3533C-XC The controller, developed by Guilin Fengde Control Technology Co., Ltd., is a low-voltage permanent magnet synchronous motor controller specifically designed for electric tricycles. It employs advanced domestic and international motor vector control algorithms to achieve precise torque control across a wide speed range, resulting in a more stable and smooth driving experience. Furthermore, it utilizes several industry-leading safety technologies to ensure the vehicle's safety and reliability under various operating conditions.

2.2 Features

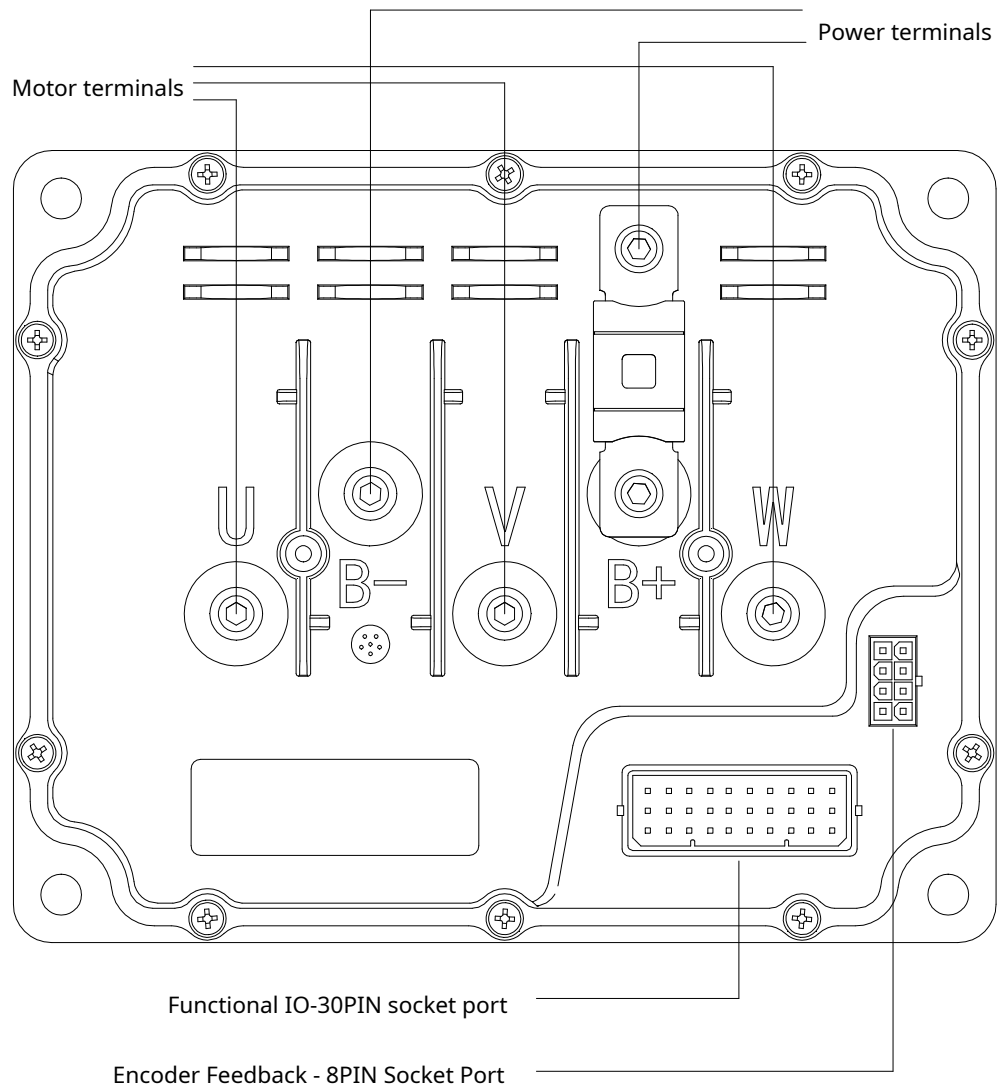
- ◇ The motor uses a high-temperature resistant, non-demagnetizing, strong vibration resistant, and dust and oil resistant encoder, which makes the motor highly reliable and eliminates concerns about damage to the motor encoder due to overheating or water ingress.
- Both the motor and controller have built-in highly reliable temperature sensors, combined with industry-leading overheat protection algorithms, which can protect the motor and controller from damage while ensuring personnel safety under extreme working conditions, further improving vehicle reliability and significantly reducing vehicle failure and repair rates.
- ◇ Supports advanced anti-rollback function. When pulling heavy objects uphill, releasing the accelerator can prevent the vehicle from losing speed or control, further improving safety and ensuring vehicle safety.
- ◇ Supports electronic speed limiting on downhill slopes to prevent vehicle stalling, making it safer to drive downhill when overloaded, while the recovered energy can increase the driving range.
- ◇ The vehicle speed can reach 40-55km/h.
- ◇ With strong torque and a load capacity of 1 ton, it can easily climb mountain roads with a 30% gradient.
- ◇ High efficiency, up to 94%, increasing vehicle mileage.
- ◇ Soft start algorithm, smooth and seamless start-up, with no jerking sensation.
- ◇ Advanced anti-tooth-smashing algorithm can significantly improve the service life of the gearbox.
- ◇ Utilizing an industry-leading field-weakening algorithm, it supports three-speed adjustment, increasing top speed by 8 km/h and providing a more comfortable driving experience.
- ◇ Supports regenerative braking function, effectively increasing driving range.
- ◇ Supports CAN communication protocol, enhancing vehicle intelligence.

2.3Key technical parameters of the controller

model		FDK3533C-XC
base Book 参 number	External dimensions (mm)	(1Overall dimensions:205*165*71.5 mm(Rain cover not included)2 Overall dimensions:205*176.6*82.1 mm(Includes rain cover)
	Maximum bus current	DC 140A
	Maximum phase current	350 A rms
	Rated operating voltage	48/60/72 V DC
	Limiting power supply range	18~96 V DC
	Power control method	SVPWMcontrol
	Insulation resistance	Room temperature or low temperature ≥20MΩHigh temperature or constant humidity ≥1MΩ
	Protection level	IP65
	Heat dissipation and cooling methods	Natural heat dissipation cooling (installed in a well-ventilated location)
base Book achievement able	Compatible motor types	Permanent magnet synchronous motor
	Compatible encoder types	(1Rotary Transformer
	achievement able Description Narrative	◇ Anti-flying vehicle: - ◇ Throttle cable breakage protection: - ◇ Encoder disconnection protection: - ◇ Driver overheat protection: - ◇ Motor overheat protection: - ◇ Low-speed ramp descent: - ◇ Speed limit on ramps: - ◇ Slope prevention: - ◇ Weak magnetic acceleration: - ◇ Energy Feedback: - ◇ Flexible start: - ◇ Three speed settings: - ◇ Speed loop control: - ◇ Motor rotation direction switching: - ◇ Battery voltage switching: - ◇ Anti-theft lock motor: - ◇ Ready signal output: - ◇ P gear supported: - ◇ High brake/low brake: - ◇ CAN communication: - ◇ 485 Communication: - ◇ Pulse Instrument: - ◇ One-line communication instrument: -
	Supported instrument types	(1)CANInstrument (supports customized protocols)2Pulse meter (3One-line communication meter (supports customized protocols)
	Operating ambient temperature	- 25℃~+60℃
	Storage temperature	- 40℃~+85℃
	relative humidity	0%~100%RH(No condensation)

3 Controller Port Introduction

3.1 Controller port distribution diagram



3.2FunctionIO_30PINsocket

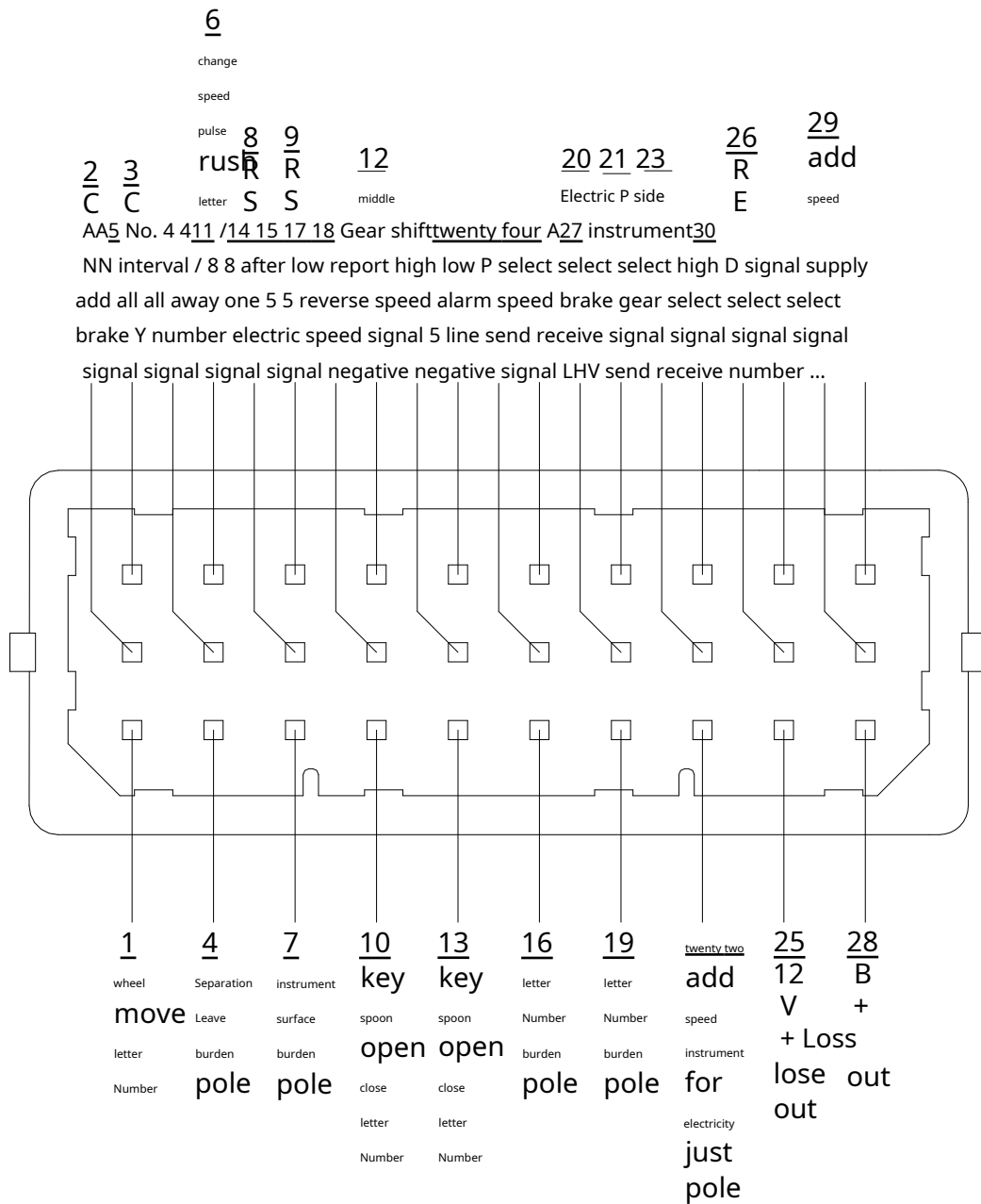
3.2.1FunctionIO_30PINSocket Port Definition Table

PINsequence	Wire harness color	port name	Port Description
A1	Brown and white	Wheel signal	Connected to an alarm to indicate that the motor is running.
A2	Red and white	CANcommunicationL	Customization based on customer needs
A3	red	CANcommunicationH	
A4	Light green and white	Isolation negative electrode	Isolation ground wire
A5	pink and white	isolation5V	isolation5VOutput reserved for Bluetooth power supply
A6	Green and white	Speed pulse signal / one-wire connection	8Pulse speed signal (5V(Level) /One-line signal
A7	black	Instrument negative terminal	Instrument ground wire
A8	Blue and white	RS485send(T+)	Debugging or command control use
A9	Purple and white	RS485take over(T-)	
A10	purple	Key switch signal	Control circuit power supply (power supply)B+enter)
A11	Brown	Back signal	Active low, back off
A12	orange	Medium/low speed signal	Active low level, toggles between medium/low speed mode.
A13	purple	Key switch signal	The alarm sends a key switch signal to the controller to activate it. controller
A14	Light blue	alarm signal	Active low level, alarm locks motor
A15	blue	high speed signal	Active low level, switches to high-speed mode.
A16	black	negative pole of signal	Low-level active signal common ground line
A17	Ash	Low brake signal	Active low, braking
A18	yellow	PSlot signal	Active low, enter/exitPFile mode
A19	black	negative pole of signal	Low-level active signal common ground line
A20	Orange and white	Voltage selection signal	Active low level, switching battery voltage level
A21	Yellow and black	PGear selection signal	Active low, enable/disablePFile mode
A22	pink	Accelerator power supply positive terminal	Output5V+Powering the accelerator
A23	black	Direction selection signal	Active low level, switches motor phase lines.
A24	gray	High brake signal	high level (12V+Effective, brakes
A25	Light green	12V+Output	Only usable at high level (12V+) Effective switch signal Number (maximum current)200mA)
A26	white	READYSignal	Output a5VLevel signal to light up the instrument READYlamp
A27	Black and white	negative pole of signal	Low-level active signal common ground line
A28	Yellow and white	B+Output	Can only be used for key switch signals (maximum current)1A)
A29	black	Accelerator power supply negative terminal	Accelerator power supply ground wire
A30	green	acceleration signal	Detect accelerometer input signal (Input range)0~5V)

Note: The wire harness colors in the table are for reference only. If you are using your own wires, the actual colors should be used. The terminals use electric vehicle controller wire harness connector plug-30P.

- Wire harness technical requirements:
- 1. The wire is free from scratches, exposed copper wires, and other defects;
- 2. 0.5 square millimeter wire, 1.6mm outer diameter, the wire must comply with RoHS;
- 3. The tensile strength between the terminal and the wire is above 6 kgf, ensuring good conductivity;
- 4. Cable: 80°C 300V;
- 5. Terminal housings must be products from designated manufacturers with an IP66 protection rating;
- 6. Terminal sleeve waterproof sleeve, flame retardant V1, temperature resistant approximately -35°C~100°C, material PVC, black, hardness 70 degrees.

3.2.2FunctionIO_30PINSocket Port Analysis Diagram



- Note: This socket has directional ports. Please check carefully during use and do not plug it in backwards!

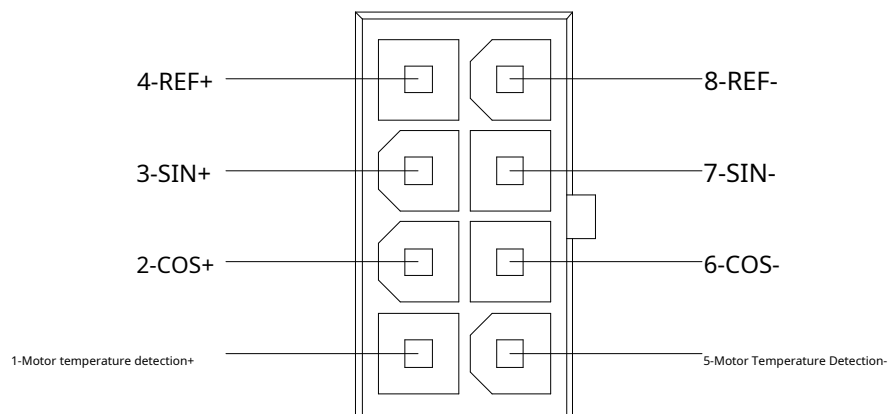
3.3 Encoder Feedback 8PIN socket

3.3.1 Encoder Feedback 8PIN Socket Port Definition Table

PIN sequence	Wire harness color	port name	Port Description
B1	Red and white	Motor temperature detection +	Used for motor temperature detection (The default temperature sensor is) KTY84-150
B2	blue	COS+	cosine signal
B3	yellow	SIN+	sinusoidal signal
B4	red	REF+	Excitation signal
B5	Black and white	Motor temperature detection -	Used for motor temperature detection (The default temperature sensor is) KTY84-150
B6	white	COS-	cosine signal
B7	green	SIN-	sinusoidal signal
B8	black	REF-	Excitation signal

- Note: The wire harness colors in the table are for reference only. If you are preparing your own wires, the actual colors should be used.
- Wire harness technical requirements:
 1. The wire is free from scratches, exposed copper wires, and other defects;
 2. 0.5 square millimeter wire, 1.6mm outer diameter, the wire must comply with RoHS;
 3. The tensile strength between the terminal and the wire is above 6 kgf, ensuring good conductivity;
- **4. Cable: 80°C 300V;**
- 5. Terminal housings must be products from designated manufacturers with an IP66 protection rating;
- 6. Terminal sleeve waterproof sleeve, flame retardant V1, temperature resistant approximately -35°C~100°C, material PVC, black, hardness 70 degrees.

3.3.2 Encoder Feedback 8PIN Socket Port Analysis Diagram



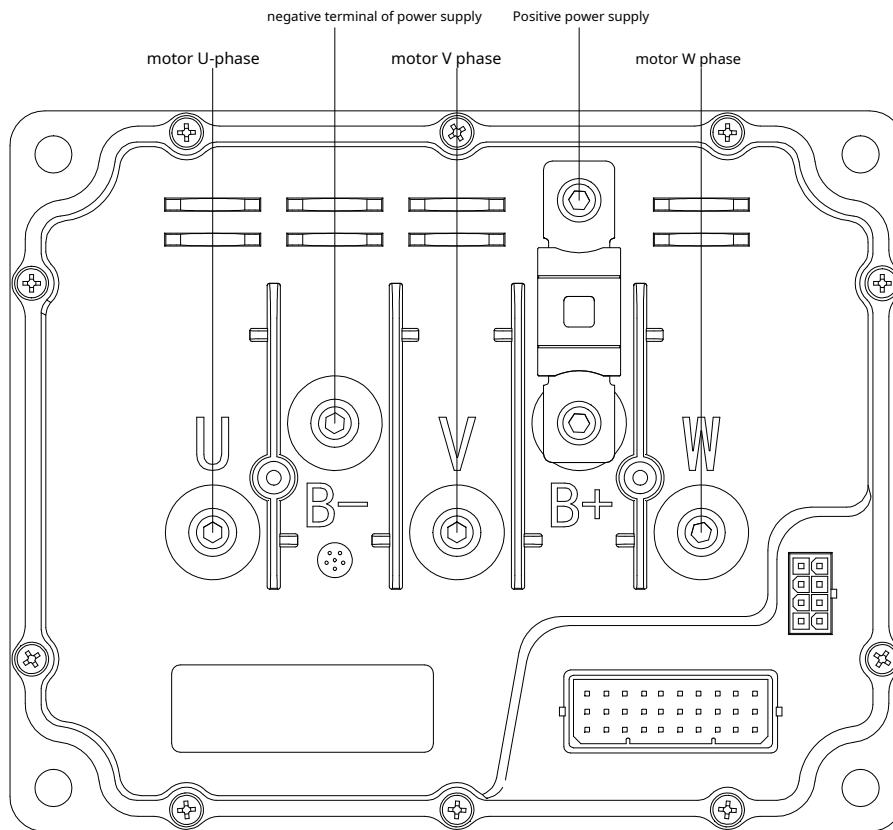
- Note: This socket has directional ports. Please check carefully during use and do not plug it in backwards!

3.4 Power supply and motor terminals

3.4.1 Power supply and motor terminal definition table

Location (Left → right)	wiring Column righteous	port name	High-temperature wear-resistant tin-plated multi-strand copper Core wire cross-sectional area	Terminal model	crimping tools
1	U	motorUMutually	25~36mm ²	Circular bare terminal RNB 38-6	ZTGYQK-70Pressing lines clamp
2	B-	negative terminal of power supply	25~36mm ²		
3	V	motorVMutually	25~36mm ²		
4	B+	Positive power supply	25~36mm ²		
5	W	motorWMutually	25~36mm ²		

3.4.2 Analytical diagram of power supply and motor terminal blocks



- Note: Do not reverse the positive and negative power supply wires!
- Note: It is recommended to use the above-defined wiring sequence for three-phase power lines.

4 Controller resource configuration introduction

4.1 signal input interface

The controller is configured with a total of 11 signal input interfaces, among which 10 high-level active digital signal input interfaces and 1 low-level active digital signal input interface. There are one analog signal input interface, and the distribution of the input interfaces is shown in the table below:

	PINsequence	port name	Trigger type	Input voltage range	Port limit voltage
letter Number lose out catch mouth	A11	Back signal	Active low (Typical value: 0V)	0~0.2V	24V
	A12	Medium/low speed signal			
	A14	alarm signal			
	A15	high speed signal			
	A17	Low brake signal			
	A18	PSlot signal			
	A20	Voltage selection signal			
	A21	PGear selection signal			
	A23	Direction selection signal			
mold Proposed letter Number	A24	High brake signal	High level active (Typical value: 12V)	10~18V	24V
	A30	Accelerator signal	(Typical value: 0~5V)	0~10V	12V

- Note: When using the above signal input interfaces, the system must operate within typical values. Exceeding the input voltage range may cause controller malfunction or damage.

- Note: The controller signal input interface is positive voltage triggered, and the reference ground is the negative power supply terminal. Do not inject negative voltage, otherwise it may cause controller malfunction or damage.

4.2 Signal output interface

The controller is configured with a total of 1 signal output interface, for 5V. The digital signal output interface is shown in the table below:

	PINsequence	port name	Output voltage value	Maximum current limit
letter Number lose out catch mouth	A26	READYSignal	5V±2%	50mA

- Note: When using the above signal output interfaces, do not use them as a power supply for other devices or equipment. Exceeding the maximum current limit may cause controller malfunction or damage.

- Note: The controller signal output interface is positive voltage, and the reference ground is the negative power supply terminal. Do not apply external voltage, otherwise it may cause controller malfunction or damage.

4.3 Voltage output interface

The controller is configured with a total of 4 voltage output interfaces, among which 2 individual 5V voltage output interface, 1 individual 12V voltage output interface, 1 individual B+ output interface. The output interfaces, including voltage output interfaces, are shown in the table below:

electricity	PIN sequence	port name	Output voltage value	Maximum current limit
Pressure	A5	isolation 5V	5V±2%	100mA
lose	A22	Accelerator power supply positive terminal	5V±2%	200mA
out	A25	12V+Output	12V±2%	200mA
catch	A28	B+Output	B+	1A
mouth				

- Note: The above voltage output interface can only be used for accelerator, Bluetooth, linear electronic brake, controller key switch signals, and other controller signals that are active high. This device is for power supply only. Do not use it to power other vehicle equipment such as lights or instruments, as this could cause excessive load and lead to controller malfunction or damage.
- Note: Do not apply external voltage to the above voltage output interfaces, otherwise it may cause controller malfunction or damage.

4.4 Communication interface

The controller is configured with a total of 3 communication interfaces as follows: CAN communication, RS485 communication and instrument communication. The communication interface distribution is shown in the table below:

	PIN sequence	port name	Communication type	Built-in terminating resistor
Pass News catch mouth	A2	CAN communication L	CAN communication	120Ω
	A3	CAN communication H		
	A4	ISO_GND		
	A8	RS485 send (T+)	RS485 communication	120Ω
	A9	RS485 take over (T-)		
	A6	Speed pulse signal / one-wire connection	Instrument Communication	-
	A7	Instrument negative terminal		

4.4.1 CAN communication

The controller has isolation CAN communication function, can be used with... CAN functional devices such as meters, BMS, VCU communication between devices, and different usage scenarios. The communication protocol needs to be customized.

- Note: Shielded twisted-pair cable is recommended. Connect two cables to each end of the bus. 120Ω termination matching resistors prevent signal reflection.
- Note: When the environment has strong interference sources, causing... CAN. When communication is intermittent or interrupted, it is recommended to connect an isolation ground wire. PIN_A4 suppressing some interference.

4.4.2 RS485 communication

The controller has isolation RS485 communication functions are generally used for controller debugging or command control.

- Note: Shielded twisted-pair cable is recommended. Connect two cables to each end of the bus. 120Ω termination matching resistors prevent signal reflection.
- Note: When the environment has strong interference sources, causing... RS485. When communication is intermittent or interrupted, it is recommended to connect an isolation ground wire. PIN_A4 suppressing some interference.

4.4.3 Instrument Communication

The controller has non-isolated one-wire communication capability and can be matched with one-wire instruments. The one-wire communication protocol for different usage scenarios needs to be customized.

(1) When matching a pulse meter, the controller sends...8Pulse speed signal (SV(Level)), needs to be connected to the negative terminal of the instrument. PIN_A7 ; (2) When matching a one-line communication instrument, the one-line communication protocol needs to be customized, and the instrument's negative wire does not need to be connected. PIN_A7

- Note: [PIN_A6] This is a multiplexed interface; its nature is determined by the program. Generally, the factory default is a speed pulse signal.

4.5 Encoder feedback interface

The controller supports resolver encoder feedback, including sine, cosine, and excitation signals. The encoder interface distribution is shown in the table below.

	PINsequence	port name	Encoder type
Editor code instrument catch mouth	B2	COS+	Resolver encoder feedback
	B3	SIN+	
	B4	REF+	
	B6	COS-	
	B7	SIN-	
	B8	REF-	

When matching a permanent magnet synchronous motor with resolver feedback, it is necessary to separately [...]B2] 、 [B3] 、 [B4] 、 [B6] 、 [B7] 、 [B8] Connect to the motor encoder feedback line;

- Note: When using the encoder feedback interface, please carefully verify the port definition to ensure a correct and reliable connection.

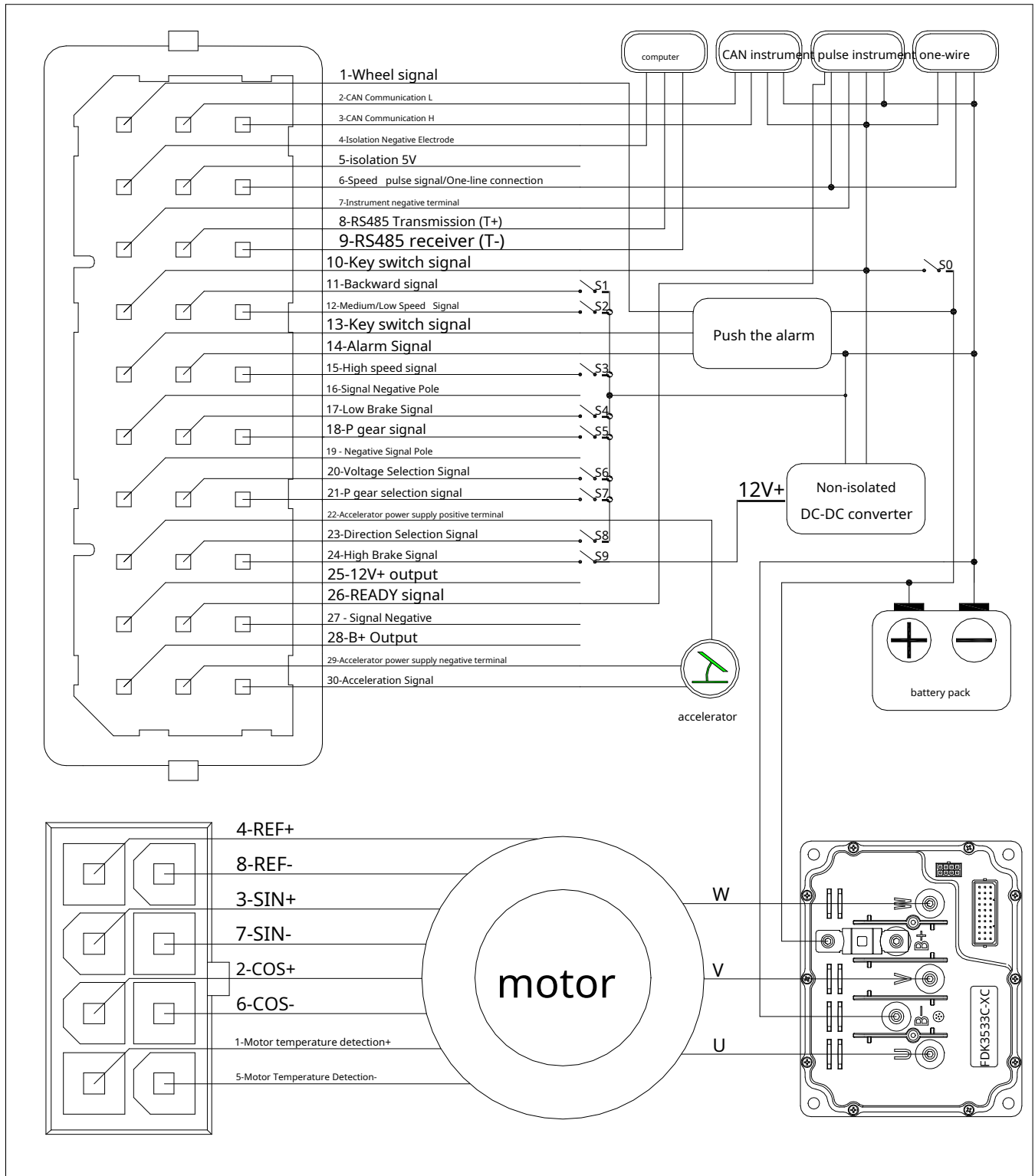
4.6 Motor temperature feedback interface

The controller supports motor temperature detection, collecting motor temperature information in real time to provide functions such as motor overheat current limiting and over-temperature protection. The distribution of motor temperature feedback interfaces is shown in the table below:

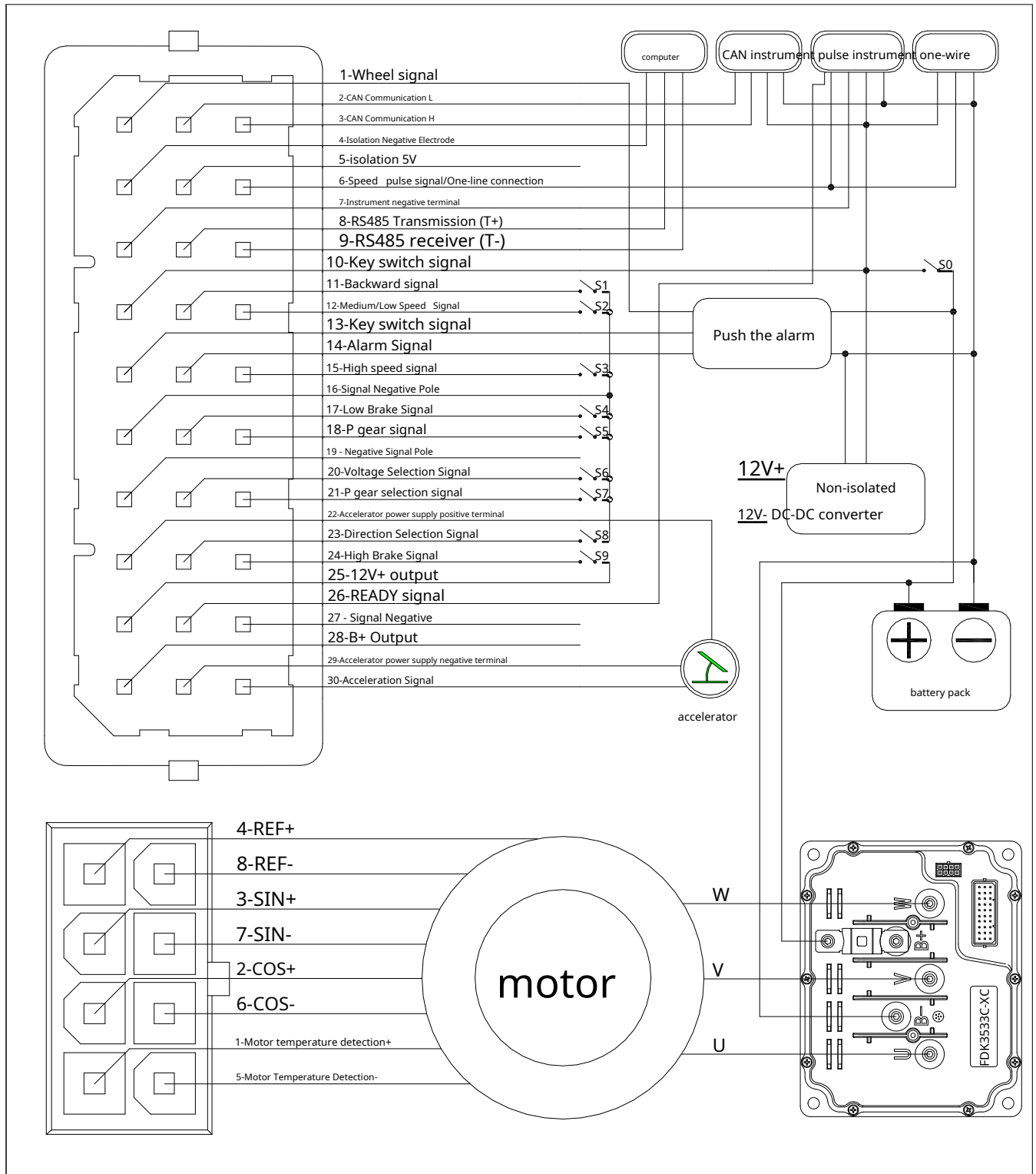
electricity machine temperature Spend opposite Feed catch mouth	PINsequence	port name	Motor temperature sensor type
	B1	Motor temperature detection +	default:KTY84-150
	B5	Motor temperature detection -	

- Note: For other types of motor temperature sensors, please inform our engineers for compatibility; otherwise, the accuracy of motor temperature data acquisition may be affected.

4.7 Typical Wiring Diagram for Controller Applications



Note: The above wiring diagram is for reference only. Actual applications require appropriate adjustments based on different scenarios.



Note: The above wiring diagram is for reference only. Actual applications require appropriate adjustments based on different scenarios.

5Install

5.1Environmental requirements

This product is an electronic product, therefore the following requirements apply to the installation environment:

- Do not install this product in environments with corrosive gases or liquids;
- Do not install this product in flammable or explosive environments;
- Do not install this product in environments with water droplets, steam, oil mist, or metallic dust;
- Do not install this product in high-heat environments;
- If you need to install this product in a vibrating environment, please contact our technical support personnel.
- Do not install this product in environments with severe electromagnetic interference or power grid fluctuations;

if you have special installation requirements, please consult our company before installation.

5.2Mechanical installation

5.2.1Security Information

Installation personnel qualifications

The controller must be installed by a professional familiar with safety and electromagnetic compatibility regulations. The installer is responsible for ensuring that the finished product or system complies with all applicable laws and regulations of the country where the controller will be used.

Pre-installation confirmation

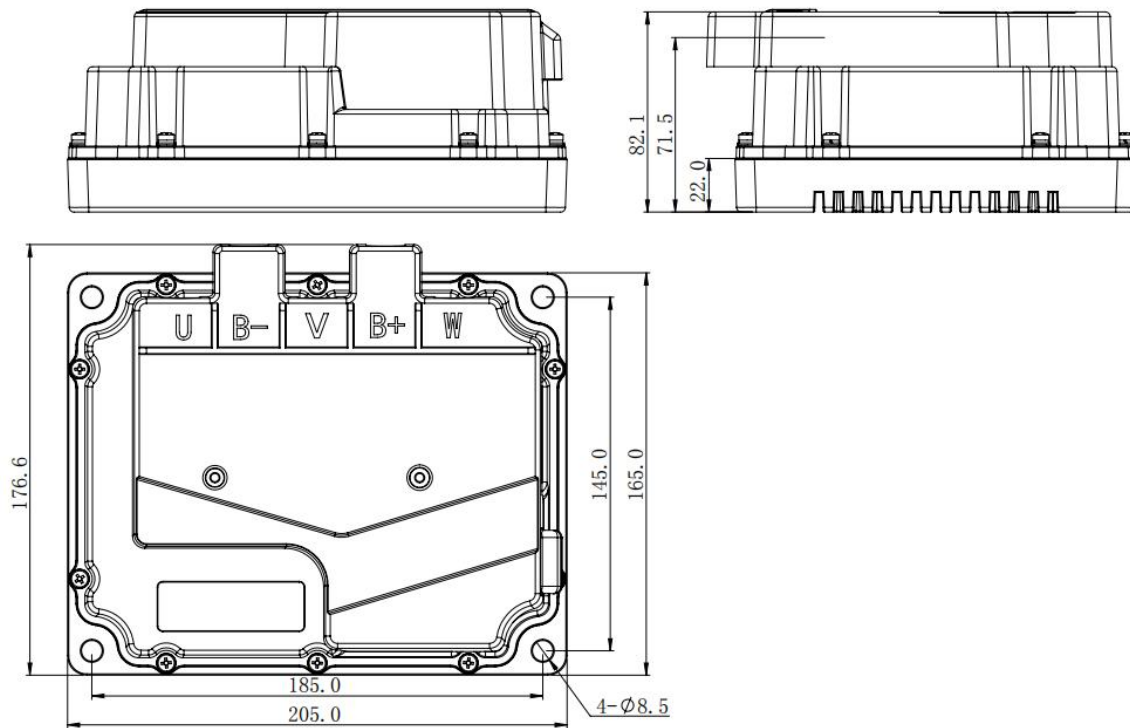
- The product model matches the ordered goods.
- The product is undamaged and all accessories are included.
- There were no foreign objects inside the product, and the screws on the outer casing were not loose.

Installation Precautions

- Do not step on the product or place heavy objects on it, as this may cause danger or damage to the product.
- Do not insert any foreign objects into the product, as this may cause danger or damage to the product.
- Please install according to the direction and spacing specified in the instruction manual; otherwise, the product may malfunction.
- Metal shavings or foreign objects that are not drilled or tapped fall into the equipment.
- Do not block the ventilation channels of the heat sink.

During installation, please do not bump the product, or allow it to fall or tip over, otherwise it may cause danger or damage to the product.

5.2.2 Controller installation dimensions diagram



5.3 Electrical installation

5.3.1 Security Information

- External wiring work for the controller can only be performed by trained and authorized qualified engineers.
- Wiring must be consistent with the wiring diagram or instruction manual.
- Do not touch the internal components of the controller, as this may cause electric shock.
- Before powering on, be sure to check the voltage rating of the controller; otherwise, it may cause personal injury and equipment damage.
- When the servo is powered on, no personnel are allowed to enter the area where the machine is moving, in order to prevent possible personal injury accidents.
- The controller housing should only be opened after all power to the system has been reliably disconnected and at least 10 minutes have been waited.
- Before use, carefully verify that the rated input voltage of the controller is consistent with the voltage of the DC power supply.
- The controller has passed the withstand voltage test before leaving the factory, and users should not conduct a withstand voltage test on the controller.
- It is forbidden to connect the power supply to the motor's power output terminals U, V, and W.
- For safety, the controller and motor must be reliably grounded.
- To facilitate input-side overcurrent protection and power outage maintenance, the controller should be connected to the power supply via an air switch or fuse switch.

5.4Post-installation check

- The controller is undamaged, securely and reliably installed, and has good heat dissipation.
- The cables are undamaged and unbroken, and the power lines, control port lines, and feedback lines are laid out separately without crossing each other.
- The wire harness interfaces are securely connected without any pulling or strain, and there is sufficient slack.
- The power supply wiring must be connected with the correct positive and negative polarities and must not be connected to other control ports or feedback ports.
- The motor phase wires have the correct phase sequence and are free of damage.
- The power line and ground line are connected correctly and reliably, with good contact.
- The input voltage to all terminals should be supplied according to the typical value, and it is strictly forbidden to exceed the allowable range.
- The encoder feedback line is correctly connected to the actual motor encoder; it is recommended to use shielded cable.

6debug

Currently, the debugging software used is "Fengdekong Debugging Assistant". For detailed usage instructions, please refer to the "Fengdekong Debugging Assistant - User Manual". If an error message appears when opening the software, such as a version mismatch, it is generally due to a missing startup configuration file. Please contact our staff to provide it.

7 Typical problems and fault analysis of controllers

A, The controller wiring is all correct, but the car doesn't move or vibrates very slowly after lightly pressing the accelerator. What could be the problem?

Answer: This indicates that the controller is starting to output power. 2 Possibility:

(1) The motor encoder is working correctly, but the motor phase sequence is incorrect. The "Swap Motor Phases" parameter in the driver needs to be changed. 1 or 0 Make the feedback speed of the test current loop positive.

(2) The motor encoder is malfunctioning, such as abnormal encoder power supply, broken wire, poor terminal contact, incorrect terminal wiring sequence, or damaged encoder.

Note: Modifying the "switch motor phase wires" parameter will only be effective if the encoder is functioning correctly.

B, The controller wiring is all correct, but after stepping on the accelerator, the car either doesn't move or the engine speeds up rapidly and it seems to run away. What could be the problem?

thing?

A: When a malfunction or overspeed occurs, the following should be done: 4 Item processing:

(1) When testing the current loop, the feedback speed must be positive and meet the following condition: $(\text{feedback speed} = \text{test current loop output frequency}) \times 60 / (\text{Number of motor pole pairs})$; if the speed feedback is negative, the parameter "switch motor phase wires" needs to be modified. 1 or 0 Make it positive.

(2) If the motor rotates but there is no speed feedback when testing the current loop, it is necessary to check whether the encoder wiring is correct or whether the encoder is abnormal and needs to be replaced.

(3) The zero bias value of the resolver's electric angle is incorrect; it needs to be corrected in (.). 1 Under the premise of recalibrating the electrical angle to zero, and calibrating to zero multiple times, the zero offset error is within... 50 Only through internal control can things function properly.

(4) The driver hardware is damaged and needs to be returned to the factory for repair.

C, Why is there no response when I turn on the key switch and step on the accelerator? Answer: In this situation, you can analyze the following aspects:

Check if the communication is normal using the host computer software, and check if the accelerator is connected correctly. The controller has the necessary output conditions.

Item:

- There is a directional signal;
- **No brake signal;**
- The accelerator has an output;
- The power supply voltage is greater than the preset (soft) undervoltage value.

When computer monitoring is not available, manual inspection is required.

8 Fault alarms and handling methods

sequence Number	Alarm methods	Representative significance	Cause Analysis
1	Long cry	High pedal malfunction	Stepping on the accelerator and turning on the key, releasing the accelerator did not resolve the problem. It is recommended to check the accelerator wiring for poor contact, incorrect wiring, or breaks. cable or damage, etc.
2	One long and two short	Precharge failure	Controller pre-charging failed; restarting the controller did not resolve the issue. It is recommended to check if the power supply is abnormal.
3	One long and three short	Overcurrent	Short circuit in the main motor circuit or encoder malfunction, such as after removing the motor and reassembling... The controller malfunction persists; it is recommended to return it to the factory for inspection.
4	One long and four short	Controller overheating	The controller temperature is too high; it is recommended to stop the vehicle and allow it to cool down.
5	One long and five short	Controller overload	The controller will report an error if the output current exceeds the rated current for a certain period of time. Overload protection is recommended; do not use if overloaded.
6	One long and six short	Short circuit fault	If the motor's main circuit is short-circuited or the current is out of control, such as after removing the motor and restarting the controller... The fault persists; it is recommended to return the device to the factory for inspection.
7	One long and nine short	Battery pack undervoltage	If the controller power supply voltage is lower than the undervoltage value, it is recommended to check the battery pack voltage.
8	One long is ten short	Battery pack overvoltage	If the controller supply voltage is higher than the overvoltage value, it is recommended to check the battery pack voltage.
9	One long and eleven short	Motor overheating	If the motor temperature is too high, it is recommended to stop the machine and allow it to cool down. If the problem persists after cooling, it is recommended to check the motor temperature feedback line.
10	One long and thirteen short	Accelerator malfunction	The acceleration signal output is abnormal; it is recommended to check whether the accelerator output voltage is correct. Within normal range
11	One long and fourteen short	Current sensor failure	Abnormal controller output current detection, such as after removing the motor and restarting the controller. The fault persists; it is recommended to return the device to the factory for inspection.
12	One long and fifteen short	Encoder error	The motor encoder is incompatible with the controller, or the encoder signal is abnormal. If there are issues such as poor contact, broken wires, short circuits, or encoder damage, please refer to the following: It is recommended to check the encoder.
13	One long and sixteen short	Communication timeout	Communication dropped or interrupted in command control mode